Rotor observer derivation

Supplement to `fw~est_rotor_001~1` and `fw~est_rotor_002~1` (`specs/firmware/est/rotor.md`). The design-C rotor estimator is a third-order Luenberger observer on $[\theta,\ \omega,\ T_d]^\mathsf{T}$ with the encoder angle as its one measurement. This note derives it end to end: the plant model, the observer and its error dynamics, the pole placement that gives the gains, and the two discretisations the firmware runs, prediction at the PWM rate and correction at the encoder rate.

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1. Plant

Newton on the rotor, with the two friction terms the model names and everything else lumped into one unknown torque T_d:

$$J\,\dot{\omega} = \tau - b\,\omega - T_c\,\operatorname{sgn}(\omega) + T_d, \qquad \dot{\theta} = \omega$$

τ is the torque the motor-control application commands ($K_t\,i_q$), b the viscous friction, T_c the Coulomb friction, J the rotor inertia. T_d is whatever acts on the rotor that the model does not name: a hand on the shaft, a load, a friction error.

T_d has no physics of its own, so the only thing the model can say about it is that it changes slowly compared with the observer. That is the constant-disturbance model:

$$\dot{T}_d = 0$$

It makes T_d a constant to be found, not a quantity to be predicted. The observer's integrator then tracks any constant T_d exactly, follows a slow drift with a lag of about $1/p_d$, and pays one third-order transient for a step. In Kalman terms the same model is a random walk, with the process noise on T_d setting the disturbance pole; in a fixed-gain observer that pole, p_d, is chosen directly.

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2. State-space form

The Coulomb term is nonlinear, so it cannot sit in a state matrix. It goes with the commanded torque into a known input the observer evaluates at its own estimate:

$$x = \begin{bmatrix} \theta \\ \omega \\ T_d \end{bmatrix}, \qquad v = \tau - T_c\,\operatorname{sat}(\hat{\omega}/\omega_{db}), \qquad y = \theta_m$$

with $\mathbf{sat}(x) = \mathbf{max}(-1, \mathbf{min}(1, x))$: the observer's Coulomb term is a ramp through zero, $\pm T_c$ beyond $\pm\omega_{db}$ and linear inside, so the input stays continuous where $\hat{\omega}$ is noise about zero rather than toggling by $2T_c$ with its sign.

The linear model is $\dot{x} = Ax + Bv$, $y = Cx$ with

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -b/J & 1/J \\ 0 & 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1/J \\ 0 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

Row i of A lists the coefficients of the states in the expression for $\dot{x}_i$. Row 1 says $\dot{\theta} = \omega$. Row 2 is the torque balance, $\dot{\omega} = (-b\,\omega + T_d + v)/J$, with the state-dependent terms in A and the known part in B. Row 3 is $\dot{T}_d = 0$.

The $1/J$ in position (2,3) is the entry that makes T_d observable: T_d pushes ω, ω pushes θ, so a lasting angle error can only be explained by T_d. Drop it and the third column of A is zero, and no gain can reach the third state.

Friction can be filed two ways. For the error dynamics in section 3 it must stay in A, because the plant has $b\,\omega$ and the observer has $b\,\hat{\omega}$ and the difference survives. For the prediction arithmetic in section 5 it can move into the known input, which makes A nilpotent and the discretisation closed-form. Both uses are shown below. The rule: A must hold every state-dependent term when analysing the error, and may hold whatever is convenient when computing the prediction.

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3. Observer and error dynamics

The Luenberger observer is a copy of the plant driven by the same known input, plus a correction proportional to the measurement residual:

$$\dot{\hat{x}} = A\,\hat{x} + B\,v + L\,(y - C\,\hat{x}), \qquad L = \begin{bmatrix} l_1 \\ l_2 \\ l_3 \end{bmatrix}$$

Subtract it from the plant with $e = x - \hat{x}$ and $y = Cx$:

$$\dot{e} = Ax + Bv - A\hat{x} - Bv - LC(x - \hat{x}) = (A - LC)\,e$$

The $B v$ terms cancel because both sides see the same input. That is the whole reason the observer is fed the commanded torque: the estimation error does not depend on what the motor is doing. The Coulomb term does not cancel exactly, since the plant has $\mathbf{sgn}(\omega)$ and the observer \operatorname{sat}(\hat\omega/\omega_{db}); the leftover $-(T_c/J)\,(\mathbf{sgn}\,\omega - \mathbf{sat}(\hat\omega/\omega_{db}))$ is zero once $|\hat\omega|$ is past ω_{db} with the right sign, and inside the band a torque of at most T_c that the observer sees as a disturbance and $\hat T_d$ absorbs.

The error is governed by $M = A - LC$. LC is an outer product whose only non-zero column is the first, because C picks out θ:

$$A - LC = \begin{bmatrix} -l_1 & 1 & 0 \\ -l_2 & -b/J & 1/J \\ -l_3 & 0 & 0 \end{bmatrix}$$

The gains sit down the first column, one per state, each in the row of the state it corrects.

For a linear system $\dot e = Me$ the trial solution $e = e_0 e^{st}$ gives $(sI - M)\,e_0 = 0$, which has a non-zero solution only where the matrix is singular:

$$\det(sI - M) = \det(sI - A + LC) = 0$$

The roots s are the eigenvalues of M, and $e(t)$ is a sum of e^{st} terms, so they are the rates at which the estimation error decays. The same statement in the Laplace domain is $(sI - M)\,E(s) = e(0)$, whose inverse puts $\det(sI - M)$ in every denominator. The roots of that determinant are the observer's poles, and choosing L is choosing where they sit.

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4. Pole placement

Write out the matrix whose determinant is the characteristic polynomial:

$$sI - A + LC = \begin{bmatrix} s + l_1 & -1 & 0 \\ l_2 & s + b/J & -1/J \\ l_3 & 0 & s \end{bmatrix}$$

Expand along the first row. The (1,1) cofactor is $(s + b/J)\,s$, the (1,2) cofactor is $-(l_2 s + l_3/J)$ with the row's -1 in front of it, and the (1,3) entry is zero:

$$\det = (s + l_1)(s + b/J)\,s + l_2\,s + l_3/J = s^3 + \left(l_1 + \tfrac{b}{J}\right)s^2 + \left(l_1\tfrac{b}{J} + l_2\right)s + \tfrac{l_3}{J}$$

Each coefficient is linear in one gain, l_1 in s^2, l_2 in s, l_3 in the constant, so any cubic is reachable. That is observability in disguise: without the $1/J$ link from T_d to ω, l_3 would not appear and the constant coefficient would be stuck.

A polynomial is fixed by its roots, so choose where the error poles should be and write down the polynomial that has them: a complex pair from the standard second-order form, natural frequency ω_n and damping ζ, plus one real pole at $-p_d$ for the disturbance state.

$$(s^2 + 2\zeta\omega_n s + \omega_n^2)(s + p_d) = s^3 + (2\zeta\omega_n + p_d)\,s^2 + (\omega_n^2 + 2\zeta\omega_n p_d)\,s + \omega_n^2 p_d$$

This desired polynomial says nothing about the plant. It is purely how fast the error should die. At $\zeta = 1$ the pair is a double real root at $-\omega_n$.

Two monic cubics are equal when their coefficients agree term by term. Equate them, one equation per power of s:

$$\begin{aligned} s^2: &\quad l_1 + b/J = 2\zeta\omega_n + p_d \\ s^1: &\quad l_1\, b/J + l_2 = \omega_n^2 + 2\zeta\omega_n p_d \\ s^0: &\quad l_3/J = \omega_n^2 p_d \end{aligned}$$

Each equation introduces exactly one new unknown, so solve top down:

$$l_1 = 2\zeta\omega_n + p_d - \frac{b}{J}, \qquad l_2 = \omega_n^2 + 2\zeta\omega_n p_d - l_1\frac{b}{J}, \qquad l_3 = J\,\omega_n^2\,p_d$$

These are the gains in `fw~est_rotor_001~1`. The J on l_3 is what makes $\hat{T}_d$ a torque in newton-metres; without it the third state would be an acceleration. The b/J terms are the friction correction: on this board b/J is 4.5 rad/s against $2\zeta\omega_n + p_d$ of 785 rad/s, so leaving them out would put the poles 0.6 % off, and they cost two terms at init.

Check against the two-state observer that preceded design C. Drop T_d and p_d: A is 2×2, the desired polynomial is $s^2 + 2\zeta\omega_n s + \omega_n^2$, the determinant is $s^2 + (l_1 + b/J)\,s + (l_1 b/J + l_2)$, and matching gives $l_1 = 2\zeta\omega_n - b/J$, $l_2 = \omega_n^2 - l_1 b/J$. Every design-C gain reduces to its design-B value as $p_d \to 0$.

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5. Prediction step at the PWM rate

The firmware runs the observer's two halves on two clocks. The prediction, $A\hat{x} + Bv$, advances the state every PWM period $T = 50\ \mu s$ in the bridge callback. The correction runs at the 1 ms encoder tick (section 6).

For one state with the input held over a step, $\dot{x} = ax + bv$ has the solution

$$x(t+T) = e^{aT}\,x(t) + \left(\int_0^T e^{a\sigma}\,d\sigma\right) b\,v$$

The first term is the state evolving on its own, the second the input accumulated over the step, each slice decayed by the time remaining after it. With $a = 0$ it reduces to $x + Tbv$, plain integration. The matrix case has the identical form:

$$x(t+T) = e^{AT}x(t) + \left(\int_0^T e^{A\sigma}\,d\sigma\right)Bv = F\,x(t) + G\,v$$

with the matrix exponential and its integral defined by the same power series as the scalar ones:

$$e^{AT} = I + AT + \tfrac{1}{2!}A^2T^2 + \tfrac{1}{3!}A^3T^3 + \cdots, \qquad \int_0^T e^{A\sigma}\,d\sigma = IT + \tfrac{1}{2!}AT^2 + \tfrac{1}{3!}A^2T^3 + \cdots$$

In general the series does not terminate. For this plant it does, once friction is moved into the known input, $v = \tau - b\,\hat{\omega} - T_c\,\mathrm{sat}(\hat{\omega}/\omega_{db})$, so that A is pure kinematics plus the unknown torque:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1/J \\ 0 & 0 & 0 \end{bmatrix}, \qquad A^2 = \begin{bmatrix} 0 & 0 & 1/J \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad A^3 = 0$$

Row 1 of A^2 is row 1 of A applied to A, which picks out row 2 of A; row 2 picks out $(1/J)$ times row 3, which is zero. Physically, A maps ω into $\dot{\theta}$ and T_d into $\dot{\omega}$, A^2 chains them into T_d into $\ddot{\theta}$, and there is no third link. A matrix like this is nilpotent, the signature of a chain of integrators, and the series stops at A^2.

$$F = I + AT + \tfrac{1}{2}A^2T^2 = \begin{bmatrix} 1 & T & T^2/(2J) \\ 0 & 1 & T/J \\ 0 & 0 & 1 \end{bmatrix}$$

θ advances by ωT plus half T_dT^2/J, the constant-acceleration kinematics; ω advances by T_dT/J; T_d is carried unchanged, the discrete face of $\dot{T}_d = 0$.

$$G = \left(IT + \tfrac{1}{2}AT^2 + \tfrac{1}{6}A^2T^3\right)B, \qquad IB = \begin{bmatrix} 0 \\ 1/J \\ 0 \end{bmatrix}, \quad AB = \begin{bmatrix} 1/J \\ 0 \\ 0 \end{bmatrix}, \quad A^2B = 0$$

$$G = \begin{bmatrix} T^2/(2J) \\ T/J \\ 0 \end{bmatrix}$$

The known torque acts on θ and ω with the same coefficients as T_d does in F's third column, as it should: both enter the plant at the same place.

`fw~est_rotor_002~1` writes the same arithmetic in a different grouping. It keeps a bare 1 on F's T_d diagonal, puts $\hat{T}_d/J$ inside the input, and uses $G = [T^2/2,\ T,\ 0]^{\mathsf{T}}$:

$$\hat{x} \leftarrow \begin{bmatrix} 1 & T & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \hat{x} + \begin{bmatrix} T^2/2 \\ T \\ 0 \end{bmatrix} u, \qquad u = \frac{\tau - b\,\hat{\omega} - T_c\,\mathrm{sat}(\hat{\omega}/\omega_{db}) + \hat{T}_d}{J}$$

Multiply out the θ row of both versions: exact gives $\hat{\theta} + T\hat{\omega} + (T^2/2J)\,\hat{T}_d + (T^2/2J)\,v$, the spec gives $\hat{\theta} + T\hat{\omega} + (T^2/2)(v + \hat{T}_d)/J$. Identical, and likewise for ω. The spec form factors $1/J$ once and treats $\hat{T}_d$ as one more term of the known input, which is also what the code does: one scalar u, two multiply-adds, a one-line change from the two-state update.

The one approximation is the friction term. With $-b/J$ kept in A, F picks up factors of $e^{-\beta T}$ with $\beta = b/J$: $F_{22} = e^{-\beta T}$, $F_{12} = (1 - e^{-\beta T})/\beta$, and so on. Writing $-b\,\hat{\omega}\,T/J$ inside u is their first-order truncation, with a neglected term of order $(\beta T)^2/2$. At $\beta = 4.5$ rad/s and $T = 50\ \mu s$ that is 2.5×10^{-8} relative. The Coulomb term has to be handled this way regardless, since the saturation cannot sit in a linear A.

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6. Correction step at the encoder rate

The correction term $L\,(y - C\hat{x})$ runs once per 1 ms tick T_s, when a fresh encoder angle exists. Forward Euler over one tick on that term alone:

$$\hat{x} \leftarrow \hat{x} + T_s\,L\,(y - C\hat{x})$$

With $y = \theta_m$, $C\hat{x} = \hat{\theta}$, and the angle difference wrapped to $(-\pi, \pi]$ so that a $0/2\pi$ crossing is not read as a full turn of error:

$$e = \mathrm{wrap}(\theta_m - \hat{\theta}), \qquad \hat{x} \leftarrow \hat{x} + Ke, \qquad K = T_s L = T_s \begin{bmatrix} l_1 \\ l_2 \\ l_3 \end{bmatrix}$$

In scalar form, the order `fw~est_rotor_001~1` writes it so every line uses the same e:

$$\begin{aligned} \hat{T}_d &\leftarrow \hat{T}_d + l_3\,e\,T_s \\ \hat{\omega} &\leftarrow \hat{\omega} + l_2\,e\,T_s \\ \hat{\theta} &\leftarrow \mathrm{wrap}(\hat{\theta} + l_1\,e\,T_s) \end{aligned}$$

One angle error is spread across three states in fixed proportions $l_1 : l_2 : l_3$. The $\hat{\theta}$ share closes the error directly; the $\hat{\omega}$ share says that an error that keeps recurring means the rotor is faster than estimated; the $\hat{T}_d$ share says that a velocity that keeps needing correction means there is a torque the model is missing. Each state is one integration further from the measurement than the last, which is why the gains climb in powers of frequency.

Units: e is in rad, eT_s in rad·s, so $l_1 eT_s$ is rad, $l_2 eT_s$ is rad/s and $l_3 eT_s$ is Nm, matching the state each line updates.

The stability bound in the spec is a condition on the Euler step itself. Forward Euler on a continuous pole at $-p$ gives a discrete pole at $1 - pT_s$, which is stable for $pT_s < 2$ and non-oscillatory for $pT_s \le 1$. The spec takes the determinant's s^2 coefficient, the sum of the three pole magnitudes, as a conservative scalar proxy and requires

$$(2\zeta\omega_n + p_d)\,T_s \le 1$$

which is 0.785 at the acceptance configuration. The exact discrete poles of the full two-rate scheme differ slightly from both the Euler and the e^{-pT_s} values because the prediction and the correction are not applied at the same instant, but with all three poles well under $1/T_s$ nothing in the acceptance depends on the difference.

What has not changed from design B: drop p_d and the third state and the correction is the two-state one with the small friction correction on the gains. The only new line is the $\hat{T}_d$ update.

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7. Numbers at the board's configuration

At the configuration the specs' acceptance bullets use, $\omega_n = 2\pi \cdot 50$ rad/s, $\zeta = 1$, $p_d = 157$ rad/s, with the bench-fit inertia $J = 1.6 \times 10^{-4}$ kg·m^2 and viscous friction $b = 7.2 \times 10^{-4}$ Nm·s/rad ($b/J = 4.5$ rad/s). The $b = 0$ column is what the acceptance tests see.

QUANTITY	SYMBOL	$b = 0$	$b = 7.2 \times 10^{-4}$	UNIT
Error pole pair	$-\omega_n$ (double)	−314	−314	rad/s
Disturbance pole	$-p_d$	−157	−157	rad/s
Angle gain	l_1	785	781	rad/s
Velocity gain	l_2	197 000	193 800	rad/s^2
Torque gain	l_3	2 479	2 479	Nm/rad
Angle correction per tick	$l_1 T_s$	0.785	0.781	rad per rad
Velocity correction per tick	$l_2 T_s$	197	194	rad/s per rad
Torque correction per tick	$l_3 T_s$	2.48	2.48	Nm per rad
Stability proxy	$(2\zeta\omega_n + p_d)\,T_s$	0.785	0.785	
Euler discrete pole, ω_n	$1 - \omega_n T_s$	0.686		
Exact discrete pole, ω_n	$e^{-\omega_n T_s}$	0.730		
Euler discrete pole, p_d	$1 - p_d T_s$	0.843		
Exact discrete pole, p_d	$e^{-p_d T_s}$	0.855		

A 1° angle error moves $\hat{T}_d$ by 0.043 Nm in one tick. The acceptance ramp of 50 rad/s^2 corresponds to a T_d of $J \cdot 50 = 0.008$ Nm, so $\hat{T}_d$ settles on it from a few hundredths of a degree of sustained error, which is why that bullet asks for e within 0.01° once settled.

8. Symbols

SYMBOL	MEANING	UNIT	WHERE IT IS SET
θ, ω	Rotor mechanical angle and velocity	rad, rad/s	plant
T_d	Unmodelled torque on the rotor, positive in the sense of positive ω	Nm	plant, estimated
τ	Commanded torque, $K_t\, i_q$	Nm	motor-control application, `fw~mc_016~1`
J	Rotor inertia	kg·m²	config, `fw~est_rotor_002~1`
b	Viscous friction	Nm per rad/s	config, `fw~est_rotor_002~1`
T_c	Coulomb friction, applied as $T_c\ \mathrm{sat}(\hat{\omega}/\omega_{db})$	Nm	config, `fw~est_rotor_002~1`
ω_{db}	The speed at which the Coulomb ramp reaches $\pm T_c$	rad/s	config, `fw~est_rotor_002~1`
θ_m	Encoder angle	rad	`fw~est_encoder_004~1`
x, $\hat{x}$, e	State, its estimate, the estimation error $x - \hat{x}$	mixed	
v, u	Known input: $v = \tau - T_c\ \mathrm{sat}(\hat{\omega}/\omega_{db})$ in Nm; $u = (v - b\,\hat{\omega} + \hat{T}_d)/J$ in rad/s²	Nm, rad/s²	
A, B, C	Continuous state, input and output matrices		section 2
L, $l_1 \dots l_3$	Continuous observer gains	rad/s, rad/s², Nm/rad	section 4
ω_n, ζ	Natural frequency and damping of the error pole pair	rad/s	config, `fw~est_rotor_001~1`

SYMBOL	MEANING	UNIT	WHERE IT IS SET
p_d	Disturbance pole, the error dynamics' third pole	rad/s	config, `fw~est_rotor_001~1`
T, T_{pwm}	PWM period, the prediction step	s (50 µs)	config, `fw~est_rotor_002~1`
T_s	Encoder tick, the correction step	s (1 ms)	`fw~est_rotor_001~1`
F, G	Discrete prediction matrices		section 5
K	Discrete correction gains, $T_s L$		section 6

Source: `docs/references/rotor-observer-derivation.md`, rendered by `tools/md_to_pdf.py` (marked + MathJax).